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Prompt 1

Okay, here is a diploma thesis of Lukas Navarra. We now want to transform it into an eight-page manuscript for the Humanoids conference <https://2026.ieee-humanoids.org/>. Please check the thesis and the relevant literature. We are ready to do additional work over the existing thesis - also run new experiments. There was a confound: the comparison of the two conditions was not fair, and that there were no significant results of the questionnaires. How would you suggest to redesign it for a paper with a clearer contribution?

Answer 1

Recommendation

I would **not** make the Humanoids paper primarily about “local vs. cloud LLMs.” In the thesis, that comparison is valuable as an engineering trade-off, but it is not internally clean: the two conditions changed STT, LLM, and TTS simultaneously, so any subjective difference cannot be attributed to the LLM itself. The thesis already acknowledges this limitation, and the small matched sample, alternation rather than full randomization, recruitment switch, and English-only deployment further weaken a causal user-study claim.

The stronger Humanoids paper is:

Tool-grounded, capability-aware LLM interaction for a real humanoid receptionist.

A possible title:

A Tool-Grounded LLM Receptionist on a Humanoid Robot: Design, Benchmarking, and Field Lessons

That paper can still include the local/cloud comparison, but as a **deployment profile**—latency, cost, upgradability, and data governance—not as the central HRI hypothesis.

Why this is the better contribution

The literature already has early receptionist and guide robots, including Valerie and Robovie, showing that public information-desk robots are feasible but historically brittle in dialogue (Gockley et al., 2005; Shiomi et al., 2009). ([CMU School of Computer Science](#)) ([dylanglas.com](#)) It also already has LLM-robot demonstrators, including FurChat as an LLM receptionist on Furhat and ChatGPT-style systems on Pepper (Cherakara et al., 2023; Qu et al., 2024; Herath et al., 2025). ([aclanthology.org](#)) ([mdpi.com](#)) ([Nature](#))

What the thesis adds, and what can be made publishable, is the **systematic design of a real institutional receptionist that constrains an LLM through deterministic tools, live institutional data sources, RAG, tablet feedback, gestures, and deployment-specific audio engineering**. The thesis implementation already exposes seven tools, including room directions, staff lookup, course schedule, canteen menu, current time, RAG over institutional documents, and session termination; it also argues that live endpoints are preferable to static document retrieval for volatile reception-desk information.

That is a clearer contribution than “cloud was rated slightly higher but local was faster,” because it addresses a real open problem in LLM-HRI: how to make open-ended language usable, safe, bounded, and useful in an embodied public-service role.

Proposed paper claim

A good central claim would be:

A humanoid receptionist becomes more reliable when an LLM is treated not as an unconstrained chatbot, but as a tool-selecting interaction manager grounded in live institutional services, explicit capability communication, and robot-side feedback channels.

This claim is testable without needing questionnaire significance. The primary outcomes become **task success, factual correctness, tool-selection accuracy, hallucination/refusal behavior, latency, and recovery from out-of-scope requests**. Questionnaires become secondary.

New experiments I would run

Experiment 1: Offline “ReceptionistBench” evaluation

Create a benchmark of reception-desk queries before running more user studies. This is cheap, controlled, and likely to produce strong results.

Use 80–150 queries across categories:

Category	Example task	Ground truth
Room finding	“Where is E-301?”	manually checked route table
Staff lookup	“Where can I find Dr. X?”	live faculty directory
Course schedule	“When is the next robotics lecture?”	live timetable
Canteen menu	“What is for lunch today?”	canteen endpoint
Institutional rules	“How do I apply for accommodation?”	RAG document corpus
Out-of-scope	“Can you print my ID card?”	correct refusal or handoff
Ambiguous query	“I need Jason”	clarification expected
ASR-robust variants	misspellings, Czech names, accents	expected tool + slot recovery

Compare at least three agents using **the same LLM, same prompt style, and text input**:

1. **LLM-only receptionist**: no tools, maybe short static prompt.

2. **RAG-only receptionist:** retrieval over documents, but no live tools.
3. **Tool-grounded receptionist:** live endpoint tools + RAG + structured failure handling.

Primary metrics:

- correct answer rate;
- correct tool-selection rate;
- slot extraction accuracy;
- hallucination rate;
- appropriate refusal rate;
- clarification rate for ambiguous queries;
- median text response latency;
- cost per 100 interactions.

This directly tests the thesis's important design insight: live APIs are better than RAG for volatile information, while RAG remains useful for policies and regulations. The current thesis already notes that reception-desk facts such as rooms, people, schedules, and menus are better served by live endpoints than static PDFs.

This benchmark could become a reusable contribution: **ReceptionistBench-FEE** or a more anonymized **Institutional Receptionist Benchmark**.

Experiment 2: Controlled on-robot user study

Then run a smaller but cleaner human study on Pepper.

Use **one fixed backend** for all participants—probably the better-performing cloud cascade or the best local cascade after fixes. Do not change STT/TTS/LLM across conditions.

Compare:

- **Condition A: current/free interaction interface**
Minimal idle tablet, no explicit capability chips, no proactive capability intro.
- **Condition B: capability-aware interaction interface**
Idle tablet shows capability chips: “Find a room,” “Find a person,” “Course schedule,” “Canteen,” “Faculty rules.” Pepper gives one short capability intro when approached: “I can help you find a room, a person, a course, or the canteen menu.”

This targets one of the thesis's strongest field observations: cold visitors asked outside Pepper's capabilities and the conversation degraded into apologies, while briefed visitors had smoother interactions.

Use a **within-subject design** if feasible. Each participant completes, for example, six scenario cards: three in Condition A and three in Condition B, counterbalanced across order and task type. Scenario cards preserve natural phrasing while making success objectively measurable:

“You are looking for the office of a faculty member.”

“You want to know when the next session of a course is.”

“You want to find a room.”

“You ask something Pepper cannot do.”

Primary outcomes:

- task success within 90 seconds;
- answer correctness;
- number of turns to success;
- number of clarification turns;
- number of out-of-scope attempts;
- whether the user notices/corrects ASR errors using the tablet;
- TTFA and full response latency;
- number of experimenter interventions.

Secondary outcomes:

- post-task confidence: “I got the information I needed”;
- perceived speed;
- perceived competence/warmth;
- short free-text feedback.

For questionnaires, I would reduce the burden. Godspeed is widely used and well established (Bartneck et al., 2009), but a conference paper with objective task data does not need 21 questionnaire items as the centerpiece. (link.springer.com) Consider **RoSAS-SF** for warmth, competence, and discomfort; it was designed as a short 6-item alternative for HRI studies under time constraints (Neuenschwander et al., 2025). (link.springer.com)

Experiment 3: Fair local-vs-cloud component study, only if you still want that angle

If local/cloud remains important, run it as a **component isolation study**, not as the main user study.

There are two clean options.

First, an **offline LLM-only comparison**:

- same text input;
- same tools;
- same tool schemas;
- same prompts;
- same ground-truth benchmark;
- only the LLM changes: Llama vs GPT or other cloud model.

Report:

- tool-call success;
- factual answer accuracy;
- hallucination rate;
- refusal quality;
- token latency;
- cost;
- privacy implications.

Second, an **on-robot full-cascade deployment profile**:

- present local and cloud as two complete deployment stacks;
- do not claim causal attribution to the LLM;
- report objective latency, reliability, cost, and data-governance trade-offs.

The current thesis has good material for the second option: local TTFA was 1.39 s median versus 3.30 s for the cloud cascade, while subjective ratings favored the cloud descriptively; however, because the two stacks differ in STT, LLM, and TTS, this is a cascade-level comparison.

Engineering fixes before collecting new data

These are worth doing before any additional study because they directly address observed failure modes.

1. **Replace passive recruitment with built-in capability discovery.**
Use idle tablet chips and a short spoken self-introduction. This turns a field-study confound into a paper contribution: capability communication for LLM robots.
2. **Fix the first-turn greeting bug.**
The thesis notes that Pepper played a fixed greeting before the LLM was involved, so a visitor who immediately asked a question got a greeting instead of an answer. That should be removed or replaced with a state machine that detects whether the first user utterance is a greeting or a task request.
3. **Use push-to-talk or a directional microphone.**
Background corridor speech was sometimes treated as visitor input. A push-to-talk button would also create a clean “active turn” boundary and make evaluation easier. The thesis already proposes this fix.
4. **Limit filler speech to once per user turn.**
“Let me check” repeated across tool retries sounded unnatural. This is a small implementation fix with likely large perceived-quality benefit.
5. **Make QR/session matching robust.**
Six of 26 questionnaires could not be uniquely matched to logs. For the paper, every

response should have an automatically embedded session ID, not a manually copied one.

6. **Make the local stack genuinely local if privacy is discussed.**

The thesis notes that query_search still uses OpenAI embeddings as a convenience, so the local condition is not fully local for RAG use. Replace this with a local embedding model if “data sovereignty” is a claim.

7. **Add Czech only if it becomes central to the study.**

Czech support would be valuable, but it can become a separate paper. For this Humanoids paper, multilingual support is best treated as future work unless you can evaluate it cleanly.

What to keep from the thesis

Keep the following as core material:

- real reception-desk deployment at FEE CTU;
- Pepper as the embodied platform;
- external microphone, AEC, LiveKit, tablet transcript, and gestures;
- tool-grounded institutional capabilities;
- live endpoint vs RAG distinction;
- objective latency logging;
- local/cloud engineering trade-off;
- field observations about discoverability, background speech, and first-turn behavior.

The system architecture is already detailed: Raspberry Pi, GPU server, LiveKit, WebRTC, HTTP tools, SSH+paplay playback, AEC, Weaviate, local and cloud agents, and session management. The tablet transcript and gesture channel are also important because LLM-HRI is not just voice-chat: prior work shows that embodied LLM robots raise expectations for nonverbal cues (Kim et al., 2024), and gesture studies show that co-verbal robot gestures affect anthropomorphism, likeability, and task-related performance (Salem et al., 2013). ([arXiv](#)) ([link.springer.com](#))

What to de-emphasize

I would de-emphasize these in the main paper:

- the human-vs-robot preference item;
- questionnaire differences between local and cloud;
- claims that the cloud model is “better” based on subjective ratings;
- broad claims about acceptance from 26 self-selected engineering-school participants;
- the 73% preference for Pepper, because the English-language context makes it hard to interpret.

The questionnaire data can stay, but as **descriptive field feedback**. Since the questionnaire effects are non-significant, they should not carry the contribution. Objective task success and groundedness should.

Suggested Humanoids manuscript structure

Humanoids 2026 is a good fit because the conference explicitly includes human-robot interaction, real-world humanoid applications, benchmarks/performance indicators, ethical and social deployment challenges, and long-term humanoid deployment; the CFP also states an 8-page maximum including references and allows one optional supplementary video.

(lists.robocup.org) The RAS page frames Humanoids as a core venue for humanoid robotics spanning mechatronics, perception, planning, learning, HRI, AI, cognition, and neuroscience. (ieee-ras.org)

A tight 8-page structure:

1. **Introduction, 0.75 page**
Problem: LLMs make receptionist robots conversational, but unconstrained LLMs are unreliable in public service roles. Contribution: tool-grounded humanoid receptionist + benchmark + field lessons.
2. **Related Work, 1 page**
Receptionist/guide robots; LLM-powered social robots; embodied cues; HRI evaluation. Keep this short and comparative.
3. **System, 1.5 pages**
One architecture figure. Explain only the parts that support the contribution: speech pipeline, tools, live endpoints, RAG, tablet, gestures, AEC, session logging.
4. **Benchmark Experiment, 1.25 pages**
ReceptionistBench: query categories, baselines, metrics, results table.
5. **Controlled On-Robot Study, 1.75 pages**
Participants, conditions, task cards, metrics, main results. Use objective measures first.
6. **Field Deployment and Lessons, 0.75 page**
Reinterpret the thesis deployment as a pilot/validation study. Include latency and qualitative failure taxonomy.
7. **Discussion, 0.75 page**
What tool-grounding buys, why capability discovery matters, privacy/local-cloud trade-offs, limitations.
8. **References, as compact as allowed**
Use only essential references.

Use the supplementary video to show: idle capability display, a successful room/person/schedule interaction, a RAG policy answer, barge-in/interruption, and a failure/clarification case.

Best concrete paper contribution set

I would formulate the contributions as:

1. **A deployed humanoid receptionist architecture** that integrates streaming spoken dialogue, deterministic tool calls to live institutional services, RAG over documents, tablet feedback, and robot gestures.
2. **A task-grounded evaluation protocol for LLM receptionists**, including a benchmark of room, person, schedule, menu, policy, ambiguous, and out-of-scope queries.
3. **Evidence that tool-grounding and capability communication improve receptionist-task performance**, measured by task success, answer correctness, hallucination/refusal behavior, number of turns, and latency.
4. **Field lessons from a real reception deployment**, including discoverability, microphone placement, first-turn handling, tablet usefulness, and local/cloud deployment trade-offs.

That combination is much more defensible than a non-significant questionnaire comparison.

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Prompt 2 - embodiment and gestures

How do we deal with the gestures? For example, if we run without the tools, then Pepper cannot easily make gestures. Shall this be part of the research question? Since we have an embodied receptionist, should the embodiment be exploited somehow?

Answer 2

Yes—the embodiment should be exploited. But I would **not** make “gestures” the main research question unless you redesign the system so that gestures are an independently controllable factor. Right now, gestures are partially confounded with tool use: in the thesis implementation, Pepper gestures mainly when the LLM calls a tool, so most visible gestures become “thinking” or “looking up,” while richer explanatory/offering gestures are largely absent because the reply is streamed for latency reasons. The thesis also explicitly notes that the gesture evaluation was rough because the visible gesture vocabulary was dominated by this tool-call mechanism rather than the full gesture set.

My recommendation: **make embodiment part of the contribution, but not by comparing “tools versus no tools” with different gesture availability**. That would create exactly the same kind of confound as the local/cloud comparison.

Best framing

I would frame the paper around:

Tool-grounded and embodiment-aware LLM interaction for a humanoid receptionist.

Then split the research questions:

RQ1: Does tool grounding improve receptionist-task performance?

This can be evaluated offline or on text logs: LLM-only vs. RAG-only vs. tool-grounded. No gestures needed.

RQ2: Does embodiment-aware interaction improve the usability of the receptionist?

This should keep the same tools and same LLM backend, and manipulate the embodiment/interface layer only.

That separation keeps the paper clean.

Do not let gestures depend on whether tools exist

The system should have a separate **nonverbal behavior manager** that receives dialogue-state events, not only tool calls. Then even a no-tool baseline can still gesture.

For example:

Dialogue event	Gesture available with or without tools?	Example behavior
Session starts	yes	wave / greet
User is speaking	yes	attentive posture / gaze
ASR final received	yes	small nod
LLM is thinking	yes	thinking pose, but only once
Pepper starts speaking	yes	open-hand conversational gesture
Clarification needed	yes	question gesture
Apology / cannot help	yes	mild regret / palms-down
Session ends	yes	goodbye / bow

Room direction result	tool-dependent	point left/right/upstairs/tablet
Staff lookup result	tool-dependent	gesture to tablet
Schedule/menu result	tool-dependent	gesture to tablet

This decouples **generic social embodiment** from **task-grounded deictic embodiment**.

The current thesis already has the right architectural ingredients: Pepper embodies the agent through speakers, tablet, and animated body, while HTTP carries gesture and tablet updates. The missing step is to make the body channel independent of the tool-call channel.

How to exploit embodiment meaningfully

For a receptionist, the most useful embodiment is not random co-speech gesturing. It is **interaction management and grounded reference**.

I would implement four classes of embodied behavior:

1. Capability signaling.

At idle, Pepper should make it visually obvious what she can do. For example, short wave, gaze to visitor, tablet chips saying “Find a room,” “Find a person,” “Course schedule,” “Canteen,” “Faculty rules.” This directly addresses the thesis observation that guided visitors had smoother interactions, while cold visitors asked outside Pepper’s capabilities and the dialogue degraded.

2. Turn-taking cues.

Use gaze, nods, listening posture, and a “thinking” pose to show whose turn it is. This is especially important because LLM latency is perceptually ambiguous: without a bodily cue, silence can look like failure.

3. Deictic gestures.

Use the body to point to the tablet, the corridor, entrance, stairs, elevator, or direction of Building A/E. This is where embodiment is functionally justified: a receptionist robot is located in physical space and can refer to that space.

4. Confidence/failure displays.

When the system cannot answer, the robot should not merely say “I don’t know.” It should visibly mark uncertainty or apology, then recover: “I can help with rooms, people, schedules, and menus.” This makes capability boundaries socially legible.

The literature supports this direction. Robot nonverbal behavior affects how people cognitively frame robots, including engagement, likeability, perceived intelligence, trust, and task behavior (Saunderson & Nejat, 2019) (link.springer.com). Salem et al. showed that humanoid hand and

arm gestures influence perceived humanlikeness and likeability, and their study compared speech-only, congruent-gesture, and incongruent-gesture conditions (Salem et al., 2013) (link.springer.com). Work on persuasive robots also suggests that bodily cues can affect user behavior, not only subjective ratings (Chidambaram et al., 2012) (pages.cs.wisc.edu).

Possible experimental designs

Option A: Keep gestures as a controlled feature, not a research question

This is the safest for Humanoids.

Run the main paper as:

Offline benchmark:

LLM-only vs. RAG-only vs. tool-grounded.

On-robot study:

Same LLM, same tools, same gesture policy for everyone. Compare capability-discovery interface vs. minimal interface.

Conditions:

Condition	Tools	Tablet capability chips	Gestures
Minimal embodied receptionist	yes	no	generic turn-taking gestures
Capability-aware embodied receptionist	yes	yes	generic + deictic/capability gestures

This tests embodiment where it matters: **does the embodied robot help users understand what it can do and complete tasks?**

Metrics:

- task success;
- number of out-of-scope first questions;
- number of clarification turns;
- time to first valid task;
- whether participant looks at/uses tablet;
- perceived competence/warmth;
- gesture naturalness as a secondary item.

This avoids a fragile “gestures alone improve everything” claim.

Option B: Make gestures a research question

This is publishable, but only if you isolate gestures.

Use the same backend, same tools, same tablet, same tasks. Manipulate only the gesture policy:

Condition	Description
No task gestures	idle life + gaze only, no arm gestures during answers
Generic gestures	greeting, nodding, thinking, goodbye, generic beats
Grounded gestures	generic gestures + deictic/tool-result gestures

Research question:

Do task-grounded gestures improve the perceived and objective usability of a humanoid LLM receptionist compared with generic social gestures?

This is cleaner than “gestures vs. no gestures,” because the real question is not whether Pepper should move; it is whether the movements should be **semantically grounded in receptionist tasks**.

Primary outcomes should still be objective:

- room-direction task success;
- whether users correctly identify where to go;
- whether they use the tablet information;
- number of repair turns;
- gaze/attention to tablet or corridor if you can code video;
- perceived gesture appropriateness.

The thesis already had a custom gesture item, but it was too broad: “natural and fit what she was saying.” For the paper, replace it with two or three sharper items:

- “Pepper’s gestures helped me understand what to do next.”
- “Pepper’s gestures matched the information she gave.”
- “Pepper’s gestures made it clear when she was thinking or waiting for me.”

Option C: Treat tools + gestures + tablet as an “embodied grounding package”

This is acceptable if you state it explicitly.

Condition A: LLM-only voice receptionist.

Condition B: embodied grounded receptionist with tools, tablet, and gestures.

But then the claim is not “tools improved performance” or “gestures improved performance.” The claim is:

An embodied grounding package improves receptionist interaction compared with an unconstrained voice-only LLM robot.

This is a broader systems claim. It may be attractive for Humanoids, but it is less analytically clean.

Technical implementation I would use

Add a small module:

Dialogue events → Nonverbal behavior manager → Pepper animation / gaze / tablet

It should listen to events such as:

session_started
user_speech_started
user_speech_ended
asr_final
llm_generation_started
tool_call_started
tool_call_completed
tts_started
tts_sentence_boundary
clarification_needed
answer_failed
session_ended

Then define gesture policies:

generic_policy:
 session_started → greet
 llm_generation_started → thinking, max once per turn
 clarification_needed → question
 answer_failed → dont_know / apology
 session_ended → goodbye

grounded_policy:

room_result → point_direction + show_map/tablet
person_result → gesture_to_tablet
schedule_result → gesture_to_tablet
canteen_result → gesture_to_tablet
out_of_scope → apology + capability_prompt

This allows **no-tools**, **tools**, **generic gestures**, and **grounded gestures** to be combined experimentally.

Paper-level recommendation

For the eight-page Humanoids manuscript, I would not make gesture generation itself the core contribution. I would make the core contribution:

A grounded LLM receptionist architecture for a humanoid robot, where language, live institutional tools, tablet display, and robot body behavior are coordinated through explicit interaction-state events.

Then include one embodiment research question:

Does capability-aware embodied interaction reduce out-of-scope use and improve task completion in a public receptionist setting?

That is stronger than asking whether gestures are “natural.” It exploits Pepper’s embodiment in a task-relevant way, and it turns one of the thesis’s deployment problems—users not knowing what Pepper can do—into a clean research contribution.

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